

Magnificent mangroves

Volunteers plant new mangrove trees in Indonesia. Mangrove forests do not just capture CO₂, but also protect land from flooding and erosion and are an important home for wildlife.

CONTROLLING EMISSIONS

CARBON CAPTURE

Carbon dioxide (CO₂, see page 168) is a greenhouse gas—one of several gases whose presence in the atmosphere keeps our planet's temperature warmer than it would otherwise be. Human activities, such as burning fossil fuels, are increasing the level of CO₂, causing global temperature to rise. Scientists are exploring different ways to reduce CO₂ levels, including planting more trees, which capture carbon dioxide from the air through photosynthesis (see page 52).

**GLOBAL HEATING**

Measurements show Earth's surface temperatures are rising by about 32°F (0.2°C) every 10 years. This is having many effects on Earth's climate, such as causing Arctic and Antarctic ice to melt. This contributes to rising sea levels.

THE GREENHOUSE EFFECT

Earth is heated by radiation from the Sun, which passes through its atmosphere. Some of it escapes back into space, but some is trapped by gases in the atmosphere in a process similar to how glass in a greenhouse traps heat. This process is called the greenhouse effect. It is important for life on Earth, but if there is too much carbon dioxide in the atmosphere, too much heat is trapped, causing the Earth to heat up.

